Section-by-section rewrite plan for the current base-operator paper

A self-contained LaTeX memorandum based on the literature and ideas map

Purpose

This document gives a concrete rewrite plan for the current paper centered on the base operator

$$L_\lambda$$

obtained from an energy decomposition of the restricted Weil quadratic form, together with the full operator

$$A_\lambda = L_\lambda + C_\lambda I + K_\lambda,$$

where C_λ is a scalar shift and K_λ is finite-rank or otherwise of very low complexity. The aim is not to change the mathematical core, but to reorganize the paper so that it is more direct, more reusable, easier to audit, and better positioned for later extensions to Dedekind zeta functions, self-dual automorphic L-functions, and Rankin-Selberg settings.

The organizing principle of the rewrite is:

abstract operator theorem + verification of hypotheses from explicit-formula data.

This prevents the paper from reading like a long one-off calculation, and it cleanly separates what is arithmetic from what is functional analysis.

Global recommendations

The rewrite should be guided by the following decisions.

1. The paper should clearly distinguish the *base-operator theorem* from all later perturbative or Connes-specific consequences. The reader should never have to infer which part is already proved and which part is a forward-looking application.

2. The main line of the paper should prove the smallest robust package of results needed for the base operator: closedness, selfadjoint realization, compact resolvent, symmetry, positivity or irreducibility input, and the spectral conclusion that is currently within reach.

3. Arithmetic computations should be pushed upstream into one explicit-formula decomposition section and downstream into appendices for constants, tails, and normalization identities. They should not be scattered across the spectral sections.

4. Whenever there are two viable proofs, the shorter proof should sit in the main line and the more flexible proof should go into an addendum if it helps future generalization.

5. The paper should include a short comparison section explaining how the explicit full operator relates to the canonical selfadjoint operator attached to the closed restricted Weil quadratic

form in the Connes-Consani-Moscovici framework, and which features are new in the present decomposition approach.

1. Recommended new top-level architecture

The paper should be rewritten into the following order.

1. Introduction and theorem map.
2. Operator-theoretic setup and notation.
3. Arithmetic input: explicit formula and decomposition of terms.
4. Construction of the base quadratic form and the operator L_λ.
5. Closedness, Markov structure, and selfadjoint realization.
6. Compact embedding and compact resolvent.
7. Symmetry, positivity, irreducibility, and the ground-state package.
8. The full operator A_λ and the low-rank correction.
9. Comparison with the canonical operator and with Zeta Spectral Triples.
10. Quantitative bounds, derived inequalities, and effective estimates.
11. Extension template for other L-functions.
12. Discussion of future directions.

This order is preferable to an arithmetic-first or computation-first presentation because it lets the reader see immediately which theorem is being proved, and it groups together the arguments that will remain valid after the zeta-specific input is replaced.

2. Section 1: Introduction and theorem map

Purpose

State, in the first two pages, exactly what the paper proves for the base operator, what it proves for the full operator, and what it does *not* yet prove. The introduction should be readable without any formulas more complicated than the definition of A_λ.

Rewrite instructions

Replace a narrative introduction with a theorem-driven one. The first theorem-style items should be a short list such as:

(T1) construction of the closed quadratic form and associated selfadjoint base operator;

(T2) compact resolvent and discrete spectrum;

(T3) symmetry and the exact spectral statement proved for the base operator;

(T4) explicit decomposition of the full operator into base part, scalar shift, and low-rank perturbation;

(T5) any effective inequalities or asymptotics already obtained.

Then add a short paragraph labeled *Relation to Connes' program* explaining which part corresponds to Step 1 heuristics and which part remains open.

What should move out of the introduction

Long derivations of the explicit formula, detailed literature comparison, and most numerical commentary should move out. The introduction should contain only a small diagram of the proof strategy and a paragraph on why the decomposition is modular enough to generalize.

Addendum recommendation

No addendum is needed here. The introduction should be short, clean, and final-form.

3. Section 2: Operator-theoretic setup and notation

Purpose

Set the Hilbert-space, form, and symmetry notation once, in a way that is independent of the Riemann zeta function as much as possible.

Rewrite instructions

Create a section whose only job is to define:

- the logarithmic interval $I_\lambda = [-L, L]$ and the L^2 space on it;
- the reflection operator $Rf(u) = f(-u)$;
- the dense test-function class used in the explicit formula;
- the quadratic-form notation, domain notation, and the conventions for even and odd subspaces.

It should also contain one abstract proposition saying that once a densely defined lower-bounded closed symmetric form is identified, the associated selfadjoint operator is obtained by the representation theorem [12, 21].

Why this helps

This removes repetitive local notation from later sections and lets every subsequent proof cite a fixed setup. It also makes later generalizations easier: only the arithmetic section should need significant editing when the L-function changes.

Alternate proof note

If the paper currently contains an explicit operator-first construction and a form-first construction, keep only the form-first construction in the main text. Put the operator-first derivation in an addendum if it clarifies intuition or is useful for numerics.

4. Section 3: Arithmetic input and explicit-formula decomposition

Purpose

This section should isolate the number-theoretic source of the operator. It should explain how the restricted Weil quadratic form splits into archimedean, non-archimedean, pole, and normalization terms, without yet forcing the reader through the whole spectral analysis.

Rewrite instructions

The section should be organized as a decomposition theorem:

> For admissible test functions, the restricted Weil quadratic form equals the sum of an archimedean difference-energy term, a prime or prime-power contribution, and explicit scalar or low-rank correction terms.

Then break the proof into separate subsections:

(a) archimedean term;

(b) non-archimedean term;

(c) pole and gamma-factor contributions;

(d) final assembly.

Every formula that is used later should be normalized here once and for all. Cite Weil, Poitou, and modern explicit-formula references for the decomposition layer, rather than rediscovering the entire framework in the middle of the paper [25, 22, 14, 20].

Main-line versus appendix split

The exact finite constants, normalization checks, duplication identities for gamma factors, and prime-sum rearrangements should be moved to appendices unless they are needed immediately in the proof of closedness or compactness.

Why multiple proof modes may be needed

For ζ, the classical explicit formula is symmetric enough that the decomposition can be made very concrete. For Dedekind zeta functions or self-dual automorphic L-functions, the same section may need a more invariant formulation in terms of local factors and completed L-functions. The rewrite should therefore present the zeta case in a way that visibly abstracts: identify which terms are genuinely special to $\mathbb{Q}$ and which arise from a generic completed L-function.

Addendum recommendation

Include an addendum containing a "dictionary of local data" for future generalization: what changes in the gamma-factor terms, conductor terms, and Euler-product side when one passes from ζ to Dedekind or Rankin-Selberg settings.

5. Section 4: Construction of the base quadratic form and the operator L_λ

Purpose

This section should extract the principal part of the form and define the base object cleanly before any perturbative discussion begins.

Rewrite instructions

State an explicit theorem of the form:

> The principal part of the restricted Weil quadratic form can be written as a symmetric difference-energy form on I_λ, possibly plus a scalar normalization term. Its closure defines a lower-bounded selfadjoint operator L_λ.

The key is to keep the theorem at the level of the form, not the formal differential or integro-differential expression. If the current paper introduces L_λ by formula first and proves its form identity later, reverse that order.

What to emphasize

Emphasize the structural gains from this decomposition:

- the main energy is nonnegative or at least has a transparent coercive part;
- lower-order arithmetic corrections are now visible as bounded or finite-rank terms;
- reflection symmetry can be checked at the level of the kernel.

Addendum recommendation

If the kernel admits more than one useful representation, place the alternate representations in an addendum. This is especially valuable if one version is best for proving positivity and another is best for numerical evaluation or generalization.

6. Section 5: Closedness, Markov structure, and selfadjoint realization

Purpose

This section should be the functional-analytic heart of the paper. It should turn the explicit form into a closed form with a canonical operator and make the positivity package visible.

Rewrite instructions

Replace long custom arguments with a short theorem-proposition chain:

(i) the principal form is densely defined, symmetric, and lower bounded;

(ii) it is closable, and its closure is closed;

(iii) it satisfies the Markov property or a positivity-preserving substitute;

(iv) hence it generates a positivity-preserving selfadjoint semigroup [12, 21].

If the Markov property is only available for the principal part and not the full form, say so explicitly. That distinction is strategically important because it explains why the base operator is the right object for Perron-Frobenius style arguments.

Why this should be early

Many later arguments become shorter if the reader already knows that the operator comes from a closed form with a positive semigroup. It also prevents confusion about whether later compactness or simplicity statements are being proved for a formal operator or for the canonical selfadjoint realization.

Alternate proof recommendation

An alternate proof of closedness based on direct norm equivalence or Fourier estimates may be worth preserving in an addendum if it would adapt more easily to other local archimedean factors or to weighted variants. The main text should keep only the cleanest proof.

7. Section 6: Compact embedding and compact resolvent

Purpose

This section should isolate the compactness mechanism and present it in a reusable way.

Rewrite instructions

The main theorem in this section should be phrased as an embedding result:

> The form domain of the closed base form embeds compactly into $L^2(I_\lambda)$, hence the associated selfadjoint operator has compact resolvent and discrete spectrum.

The proof should be written in one of the following modes, ranked by preference.

1. If available, use a direct compact embedding theorem for the relevant energy space.
2. Otherwise use a Kolmogorov-Riesz style tightness and translation estimate argument on the bounded interval.
3. Only if necessary, use a more ad hoc resolvent compactness proof.

The paper should cite the general compactness framework and then keep the verification local to the base form [3, 12].

Why multiple proofs may be worth preserving

Compactness is one of the places where different future targets require different methods. For the zeta case on a bounded interval, a direct compactness proof may be shortest. For Dedekind or automorphic generalizations, weights, additional trace terms, or weaker kernel regularity may make the abstract embedding route more robust. For this reason, the paper should include one alternate compactness proof in an addendum if it is not much extra work.

Addendum recommendation

Addendum A should contain an alternate compactness proof in a style suitable for future extensions. If the main text uses Kolmogorov-Riesz, the addendum should present the form-domain embedding theorem viewpoint; if the main text uses abstract embedding, the addendum should present the direct interval argument.

8. Section 7: Symmetry, positivity, irreducibility, and the ground-state package

Purpose

This is the section where the paper should cash in the functional-analytic structure for real spectral conclusions.

Rewrite instructions

Split the section into four short subsections.

(a) Reflection symmetry and invariant even/odd subspaces.

(b) Positivity preservation of the semigroup.

(c) Irreducibility or positivity improvement.

(d) Spectral consequence: simplicity and positivity of the ground state for the base operator.

The main conceptual chain should be explicit:

closed symmetric form $\Rightarrow$ positive semigroup $\Rightarrow$ irreducible or positivity improving $\Rightarrow$ simple lowest eigenvalue.

The evenness statement should then be presented as a symmetry corollary: if the ground state is unique up to scale and the operator commutes with reflection, the positive ground state must be even.

Why this needs careful packaging

The paper should avoid mixing together three distinct notions: positivity preservation, positivity improvement, and simplicity of the first eigenvalue. They are related but not identical. The rewrite should make clear what has actually been proved and which theorem is being invoked [17, 1, 2].

Why multiple proof techniques may be needed

For the zeta base operator, irreducibility may be proved from the support properties of the jump kernel. In other settings, especially if archimedean and non-archimedean contributions are packaged differently or if the principal form is only dominated by a Dirichlet form, one may need an invariance-of-ideals argument or a domination argument instead of a direct connectivity proof. The paper should therefore include one alternate irreducibility proof in an addendum if available.

Addendum recommendation

Addendum B should present an alternate route to simplicity or irreducibility. Good candidates are:

- a semigroup-irreducibility argument using invariant sets;
- an equality-case analysis in the variational problem;
- a positivity-improving argument via strict kernel positivity.

This addendum would materially strengthen the paper's robustness.

9. Section 8: The full operator A_λ and the low-rank correction

Purpose

This section should explain exactly how the base operator and the full restricted Weil form are related, and what is already proved about the full operator as a consequence.

Rewrite instructions

Open with a precise proposition identifying the decomposition

$$A_\lambda = L_\lambda + C_\lambda I + K_\lambda,$$

and explicitly state the rank or structural complexity of K_λ whenever possible. Then separate the discussion into three layers:

(i) what the scalar shift changes and does not change;

(ii) what boundedness or finite-rankness of K_λ implies automatically;

(iii) what remains open about the bottom eigenvalue of the full operator.

This is where Kato and Birman-Schwinger language should appear [16, 13].

What the section should not do

It should not pretend that all base-operator conclusions automatically transfer to the full operator if that has not been proved. The rewrite should be honest: some properties are stable under bounded perturbation, some under small perturbation, some only under extra symmetry assumptions, and some remain future work.

Addendum recommendation

Addendum C should contain a more ambitious perturbative toolkit for future use: Kato continuation in a coupling parameter t for $L_\lambda + tK_\lambda$, a Birman-Schwinger reduction for the bottom eigenvalue, and any determinant identities natural for a rank-two perturbation. This addendum need not be used in the main theorem, but it would make the paper much more useful later.

10. Section 9: Comparison with the canonical CCM operator and Zeta Spectral Triples

Purpose

This section should explain why the present operator is not ad hoc, and how it relates to the canonical selfadjoint operator attached to the full restricted Weil quadratic form in the Connes-

Consani-Moscovici framework.

Rewrite instructions

The section should make three points.

(1) The full operator in the paper should be identified, at the level of the closed form, with the canonical selfadjoint operator attached to the restricted Weil quadratic form.

(2) The present paper contributes an explicit decomposition into principal form, scalar term, and low-rank correction, which is not the same thing as the abstract representation-theorem packaging.

(3) The Connes-Consani-Moscovici papers provide finite-section, prolate, determinant, and real-zero outputs that complement rather than replace the current analysis [5, 4, 7, 6].

This comparison section should be concise and non-polemical. It should present the two approaches as complementary.

Why it belongs in the main body

This section justifies the broader significance of the paper and clarifies to expert readers how the present work fits into the modern Connes program. It also creates a natural place to cite possible future interactions with finite-section approximations and determinant formulas.

Addendum recommendation

No long technical addendum is necessary unless you have a precise finite-section convergence estimate. If such an estimate exists, put it in Addendum D as a forward-looking proposition.

11. Section 10: Quantitative bounds, derived inequalities, and effective estimates

Purpose

Gather all effective estimates in one place so the reader can see what is qualitative and what is quantitative.

Rewrite instructions

This section should assemble the consequences of the earlier theory into quantitative statements such as:

- bounds on the lowest eigenvalue and spectral gap;
- norm bounds on the low-rank perturbation;

- prime-tail truncation bounds;
- explicit constants arising from gamma-factor identities, product formulas, or the prime number theorem;
- any trace or counting estimates that are already rigorous.

It is often better to gather these in one proposition-and-corollary section rather than distribute them across the paper.

Number-theoretic inputs to foreground

This section should explicitly advertise which classical formulas are feeding into the operator estimates:

- Euler product and logarithmic derivative identities;
- gamma-function identities, including duplication and shift formulas;
- Hadamard product and Ξ-function normalization identities;
- explicit formulas in Weil-Barner-Poitou form;
- prime number theorem type estimates and explicit prime-sum bounds [9, 23, 10, 11, 24].

This is the section where the paper can most naturally advertise how arithmetic information strengthens the operator theory.

Addendum recommendation

Put long calculations of constants, prime tails, and special-function identities into Addendum E. The main text should display only the final bounds and explain how they are used.

12. Section 11: Extension template for Dedekind zeta, automorphic, and Rankin-Selberg settings

Purpose

This section should explain how much of the argument is structural and how much is special to ζ.

Rewrite instructions

Present extension as a template, not as a claim that everything already works. A useful template theorem would say:

> Whenever a completed L-function admits an explicit-formula decomposition yielding a symmetric principal energy form on a bounded logarithmic interval together with bounded or finite-rank correction terms, the operator-theoretic core of Sections 4–8 applies verbatim once the corresponding hypotheses are verified.

Then discuss three separate cases.

(a) **Dedekind zeta functions.** Likely the easiest next target because the explicit formula is closest to the zeta case, though archimedean factors and discriminant terms must be tracked carefully.

(b) **Self-dual automorphic L-functions.** Plausible if the symmetry needed for the evenness step survives in the right form and if the local factors can still be separated into principal plus lower-order pieces.

(c) **Rankin-Selberg L-functions.** Potentially richer for estimates and trace-formula connections, but more likely to need a modified symmetry package or a replacement for literal evenness.

Cite the explicit-formula and automorphic references here rather than in the introduction [19, 18, 15, 26, 8].

Why multiple arguments may be unavoidable

Different L-functions may force different proof styles for the same property. For example:

- compactness might come from bounded-interval energy compactness in one case and from weighted compact embedding in another;
- positivity improvement may be immediate from the kernel in one case but require domination arguments in another;
- the parity conclusion may be literal evenness for self-dual zeta-type cases, but only an involutive symmetry statement in a more general automorphic setting.

The rewrite should say this openly. It makes the paper look stronger, not weaker, because it shows the authors know which parts are structural and which are model-dependent.

Addendum recommendation

Addendum F should contain a one-page extension checklist: which hypotheses must be re-verified to port the argument to a new L-function.

13. Section 12: Discussion and future work

Purpose

The conclusion should not be a generic summary. It should be a roadmap.

Rewrite instructions

End with a structured list of next targets:

1. strengthen the ground-state package for the base operator;
2. develop a perturbation theory for the full operator and test whether simplicity/evenness survives the rank-two term;
3. compare the exact operator to finite-section CCM models;
4. generalize the arithmetic verification layer to Dedekind zeta functions and self-dual automorphic L-functions;
5. mine number-theoretic formulas further for effective constants and trace identities.

This is also the right place to say that some alternate proofs are included in addenda precisely to support future generalization.

14. Recommended addenda and appendices

The paper would benefit from explicitly planned technical addenda. The following split is recommended.

Addendum	Recommended contents	Why it is worth including
A. Alternate compactness proof	A second route to compact resolvent, preferably in a form-domain embedding style if the main text uses a direct interval argument, or vice versa.	Compactness is one of the most likely places where future generalizations will need a different proof. Keeping two proofs makes the paper more reusable.
B. Alternate irreducibility or simplicity proof	Either an invariant-set proof, a positivity-improving kernel proof, or a variational equality-case proof.	This materially increases robustness and reduces the chance that the main proof depends on an unnecessarily delicate point.
C. Perturbative toolkit for A_λ	Kato continuation in t, finite-rank or Birman-Schwinger reductions, and parity-preserving perturbation lemmas.	This is the natural bridge from the base paper toward a full simple-even theorem for the canonical operator.
D. CCM and finite-section comparison notes	Precise identification of the full operator with the canonical closed-form operator, plus any rigorous comparisons to prime-truncated models.	Helps position the paper relative to Zeta Spectral Triples and prepares later work on approximation and determinant outputs.

Addendum	Recommended contents	Why it is worth including
E. Arithmetic constants and special-function identities	Duplication formulas, gamma-factor simplifications, product formulas, prime-tail bounds, and normalization checks.	Keeps the main text readable while preserving the detailed calculations needed for auditing and reuse.
F. Generalization checklist	A modular list of hypotheses to check for Dedekind, automorphic, and Rankin-Selberg variants.	Converts the paper from a one-off zeta paper into a template for a broader program.

15. What to cut, condense, or demote

A successful rewrite is not only about adding structure; it is also about removing friction. The following material should usually be shortened or moved.

1. repeated versions of the same kernel or form identity;
2. long constant computations inside major proofs;
3. literature discussions that do not immediately support a theorem or a future extension;
4. operator formulas introduced before the closed-form framework is established;
5. numerical or heuristic remarks that interrupt formal arguments.

Whenever possible, replace these with a one-sentence pointer to an appendix or addendum.

16. Minimal theorem package for the rewritten paper

If the rewrite aims for the strongest clean paper with the least risk, the main body should aim to prove exactly the following package.

(M1) the explicit decomposition of the restricted Weil quadratic form into principal energy part plus lower-order terms;

(M2) construction of the closed base form and its selfadjoint operator L_λ;

(M3) compact resolvent and discrete spectrum of L_λ;

(M4) the symmetry and ground-state theorem actually available for L_λ;

(M5) the decomposition of the full operator into $L_\lambda + C_\lambda I + K_\lambda$ with explicit control on K_λ;

(M6) quantitative inequalities and bounds that already follow rigorously.

Everything beyond this package should either be corollary material or future-work material. This is the cleanest way to make the paper easy to evaluate and hard to misread.

Bottom line

The rewrite should make the paper look less like a long derivation tied to one explicit formula and more like a reusable operator theorem with a zeta-specific verification layer. The two highest-value improvements are:

1. moving all structural operator arguments into a short, theorem-driven spine;
2. relocating alternate proofs and long arithmetic computations into clearly labeled addenda that support robustness and future extension.

If this is done well, the paper becomes simultaneously easier to referee, easier to generalize, more visibly connected to the Connes-Consani-Moscovici program, and better suited as a foundation for later work on the full operator A_λ.

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